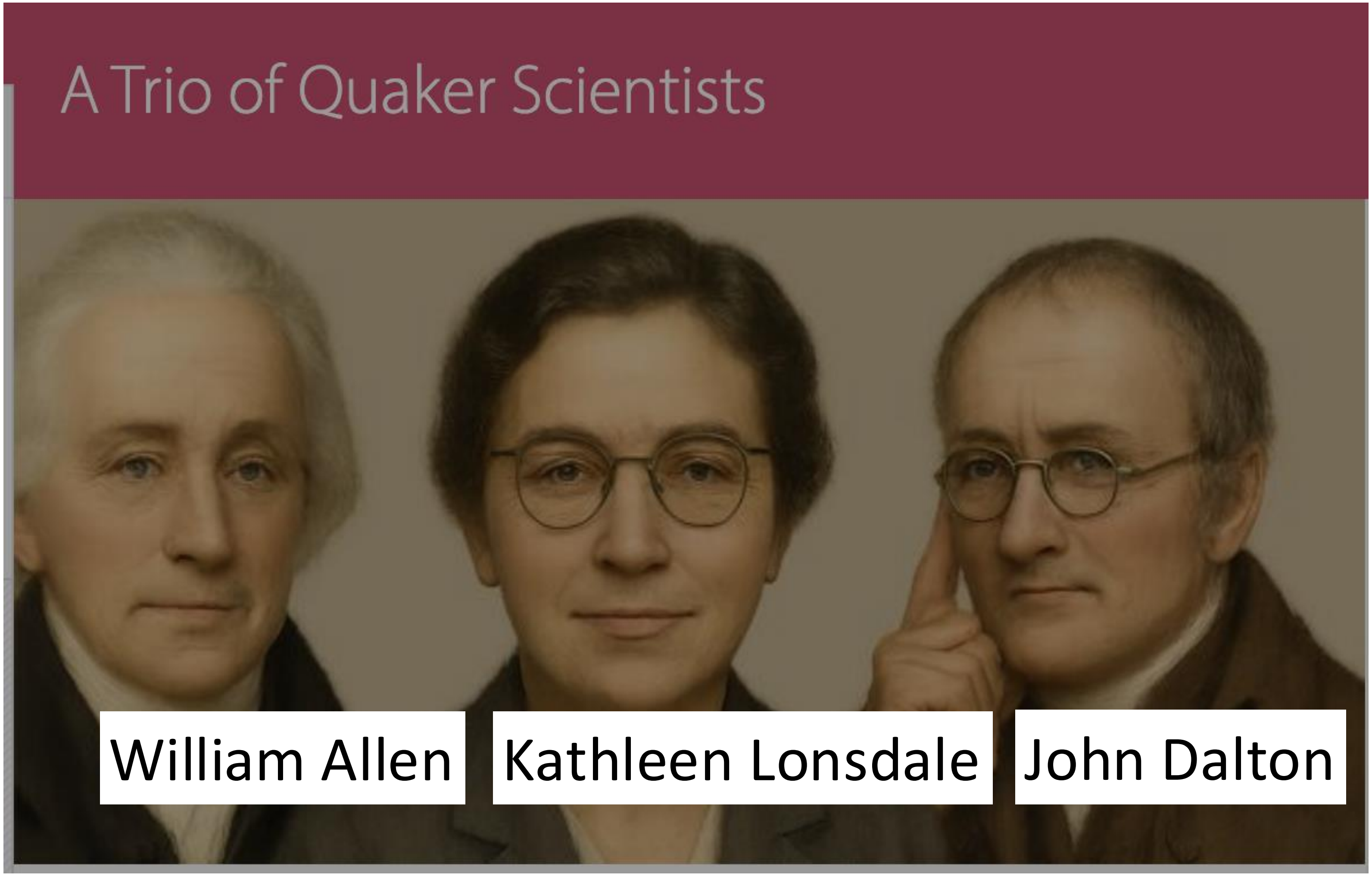


Graham Waterhouse and Peter Murray-Rust

Jesus Lane Meeting House, 2025-09-18

Early Lives

Name	**Place of birth**	**Family**	**Education**
John Dalton	Cumberland	Poor Quaker	Local Quaker School, individual mentors, self-taught
William Allen	Spitalfields, London	Merchant class Quaker	Quaker School, Rochester, mentors, self-taught
Kathleen Lonsdale	Co. Kildare, Ireland	Poor Baptist	Ilford High School, Bedford College, London

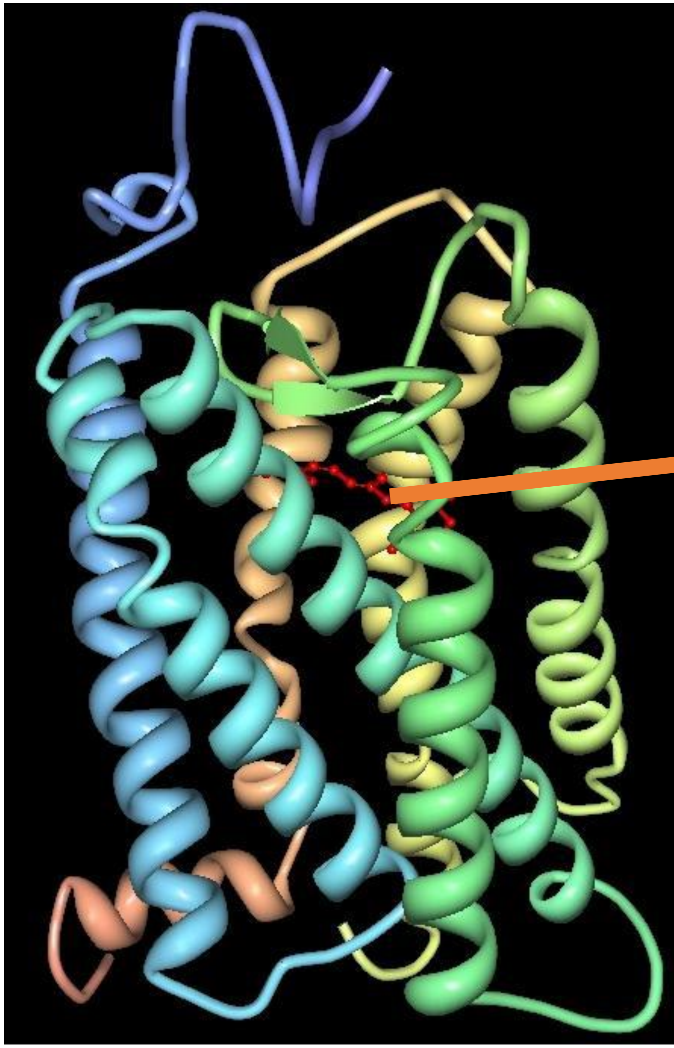
The SPICES Principles

- **Simplicity**:
- **Peace**:
- **Integrity**:
- **Community**:
- **Equality**:
- **Stewardship**:

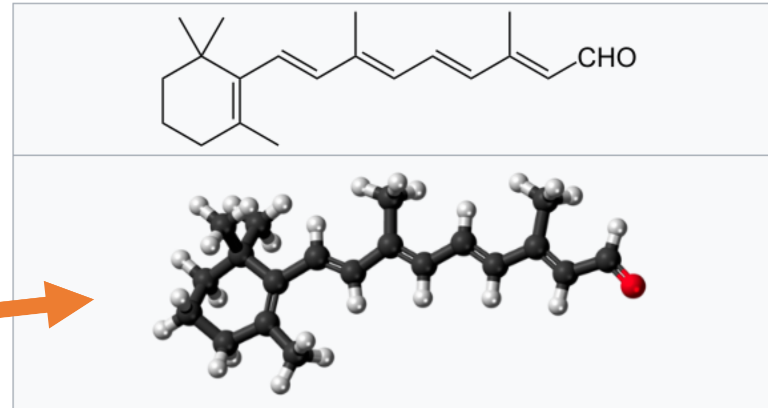
Themes:

- Science + religion
- Education, Quaker schools
- Outreach
- Dissension

The finding of one generation will not serve for the next. It tarnishes rapidly except it be reserved with an ever-renewed spirit of seeking.
Arthur Eddington



All-trans-retinal



Names

IUPAC name

Retinal

Systematic IUPAC name

(2*E*,4*E*,6*E*,8*E*)-3,7-Dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraenal

https://en.wikipedia.org/wiki/Vertebrate_visual_opsin

Three-dimensional structure of bovine rhodopsin. The seven transmembrane domains are shown in varying colors. The retinal chromophore is shown in red.
Wikipedia

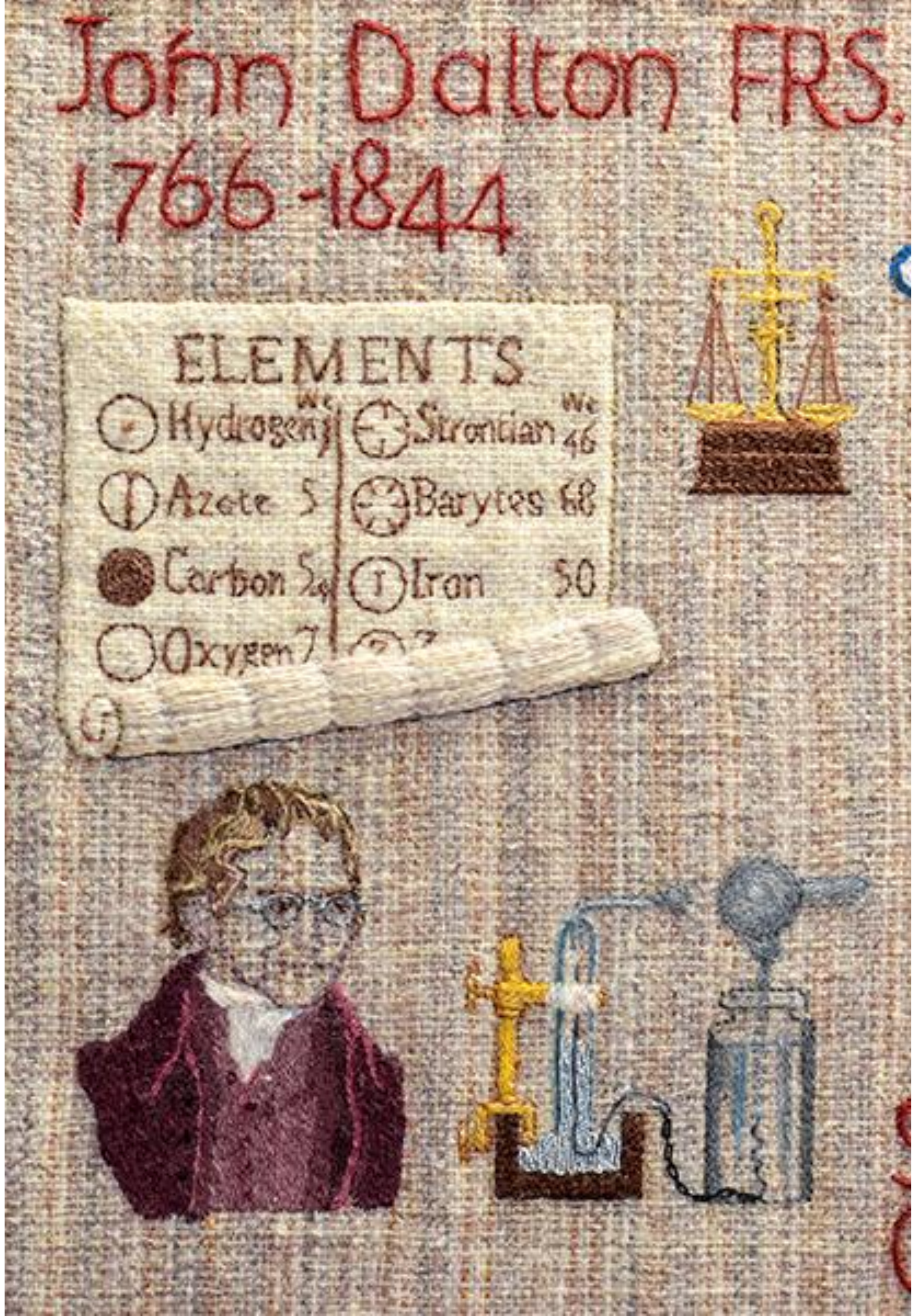
Atoms responsible for John Dalton's colour blindness

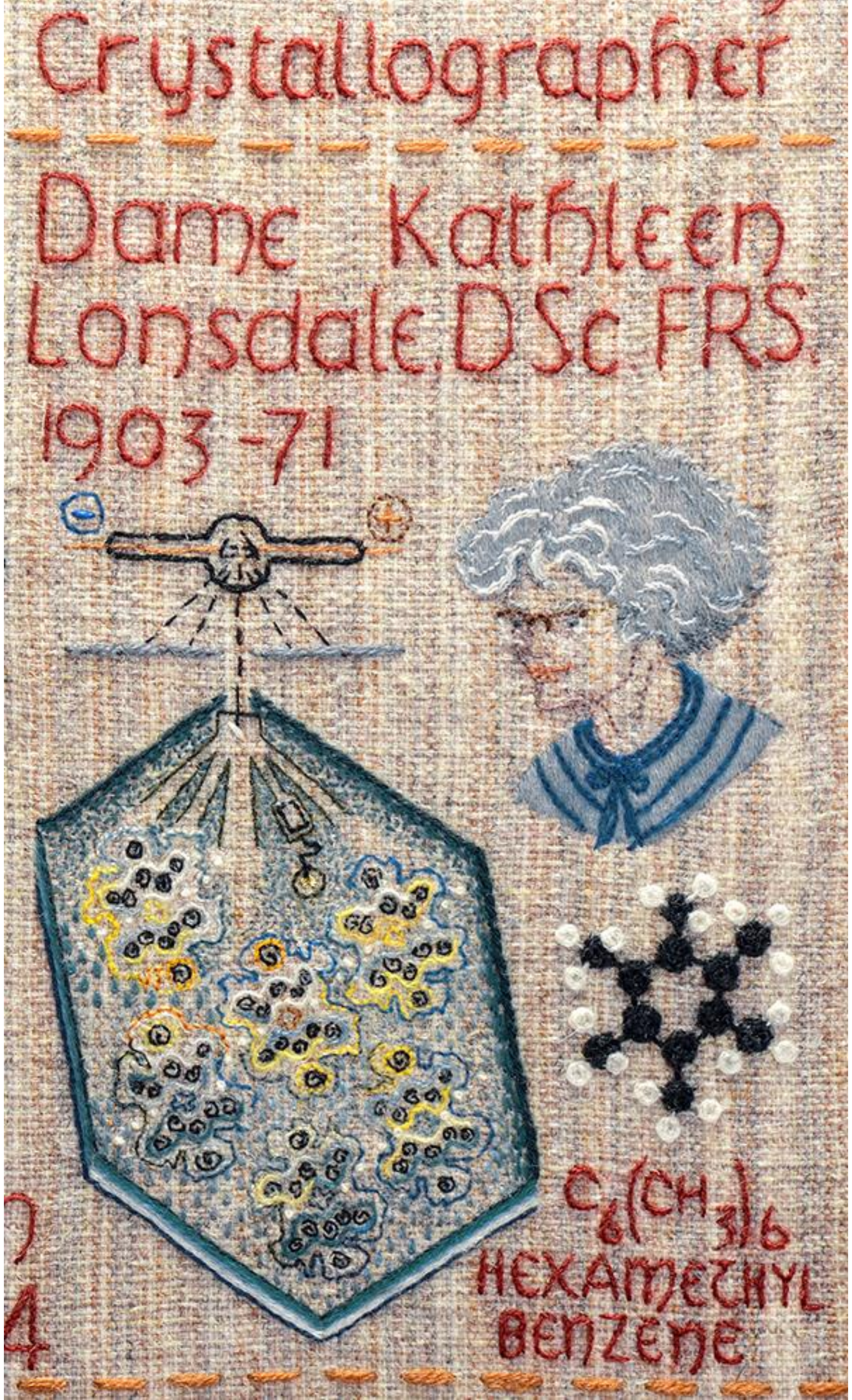
Knowledge Graphs for the scientists

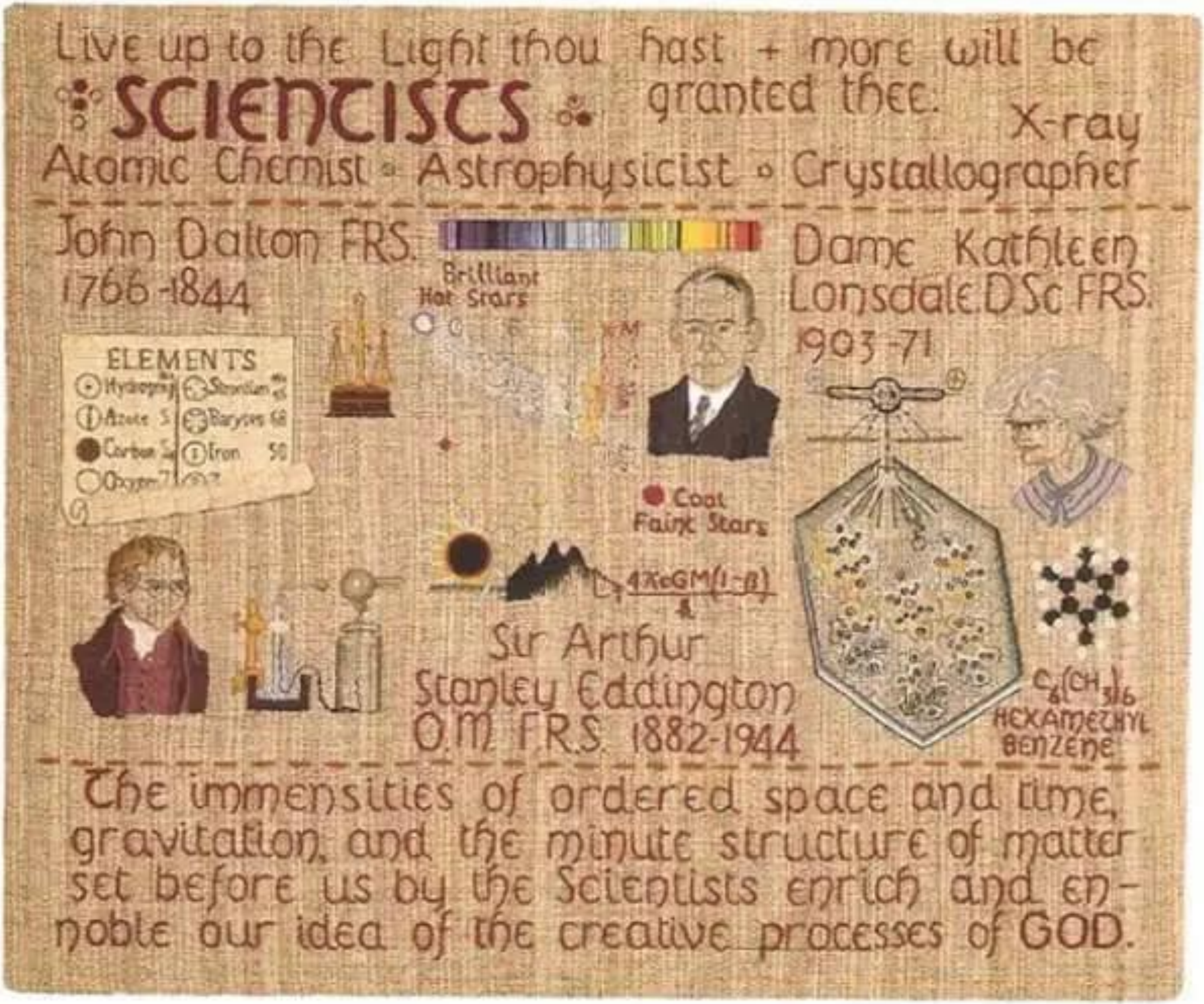
Cursor AI was used to generate knowledge graphs of relatives, mentors/mentees

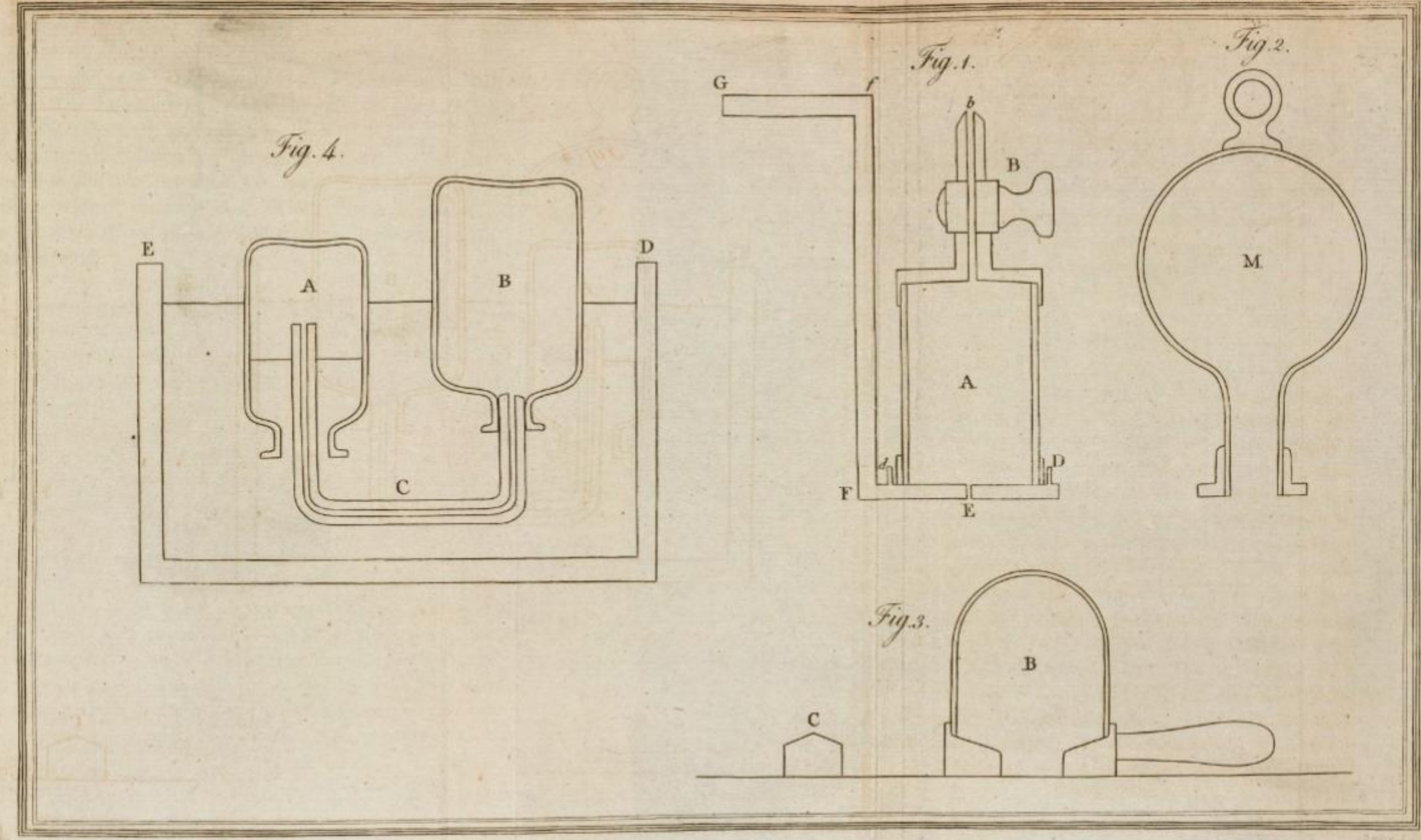
William Allen


WILLIAM ALLEN 1770-1843
Businessman + Scientist
"---cheerful, in the light around me thrown
Walking as one on pleasant service led,
Doing God's will as if it were my own,
Yet trusting not in mine
but in his strength alone"
J G WHITTIER
BRITISH AND FOREIGN SCHOOL SOCIETY
2 Plough Court
Royal Institution Lectures
Lindfield Agricultural Settlement

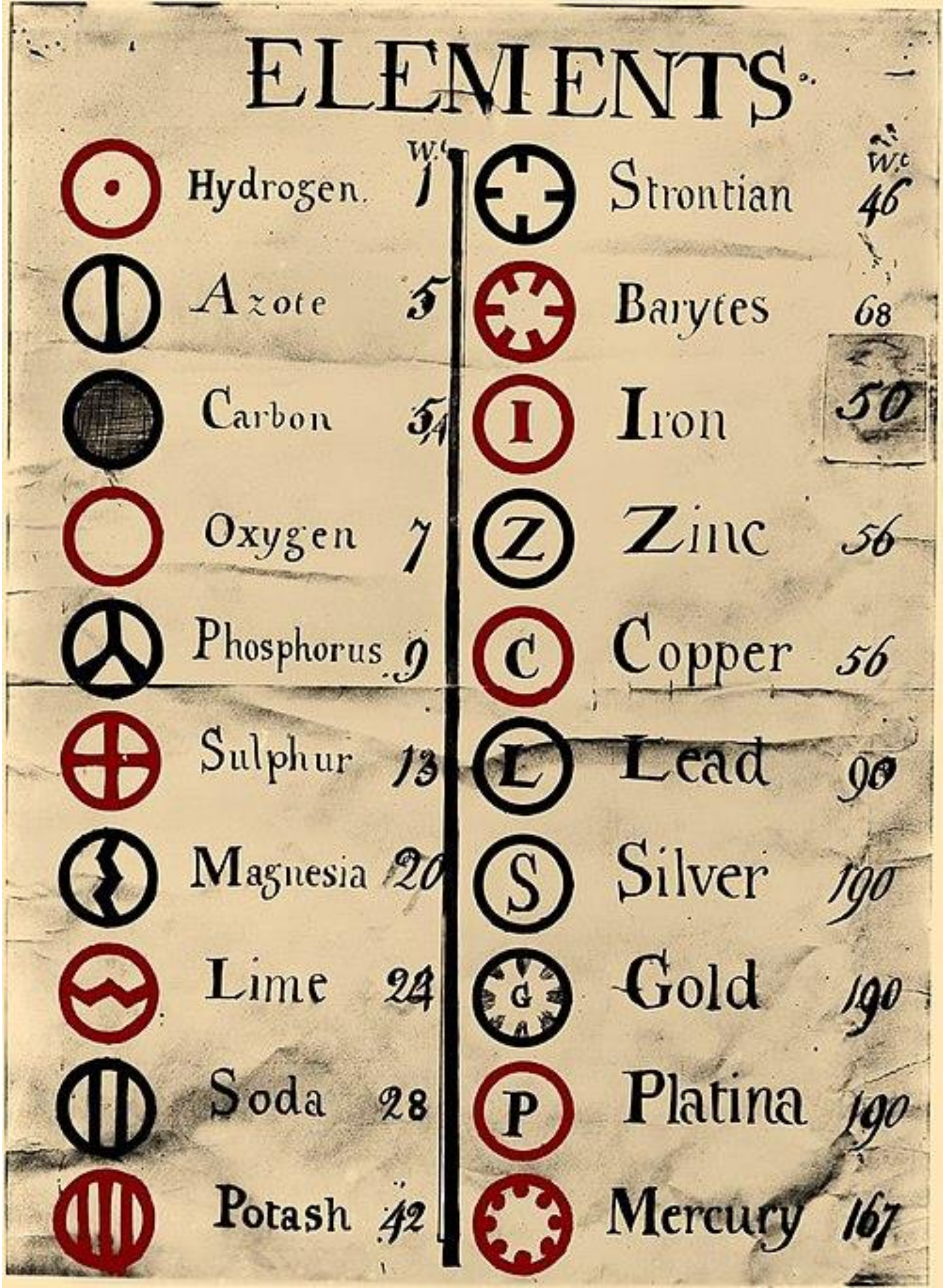

John Dalton FRS.
1766-1844
ELEMENTS
Hydrogen
Strontian 46
Azote 5
Barytes 68
Carbon
Iron 50
Oxygen 7


Crystallographer
Dame Kathleen Lonsdale, DSc FRS.
1903-71
$C_6(CH_3)_6$
HEXAMETHYL
BENZENE


Live up to the Light thou hast + more will be granted thee.
SCIENTISTS
Atomic Chemist
Astrophysicist
X-ray Crystallographer
John Dalton FRS.
1766-1844
ELEMENTS
Brilliant Hot Stars
Dame Kathleen Lonsdale D.Sc FRS.
1903-71
Coal Faint Stars
Sir Arthur Stanley Eddington O.M F.R.S. 1882-1944
C6(CH3)6
HEXAMETHYL BENZENE
The immensities of ordered space and time, gravitation, and the minute structure of matter set before us by the Scientists enrich and ennoble our idea of the creative processes of GOD.



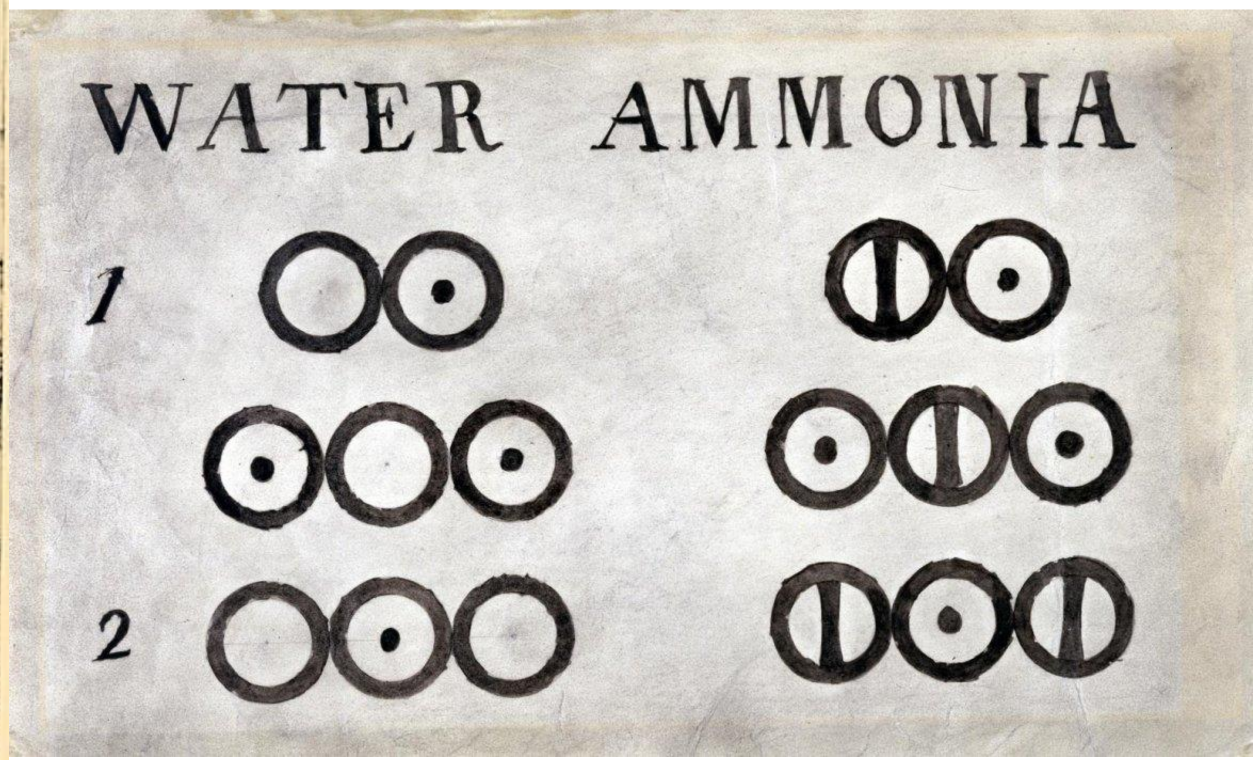
Scanned images copyright © 2017,
Royal Societyhttps://doi.org/10.1098/rstl.1783.0008open_in_new



ELEMENTS.

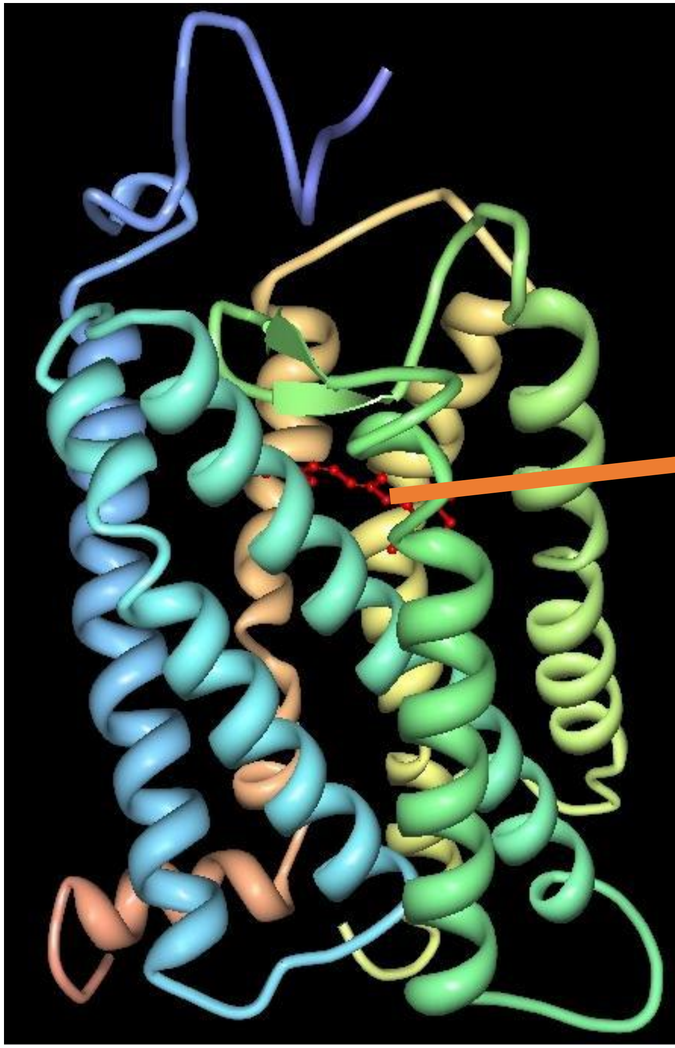
	Wt		Wt
Hydrogen.	1	Strontian	46
Azote	5	Barytes	68
Carbon	5,4	Iron	50
Oxygen	7	Zinc	56
Phosphorus	9	Copper	56
Sulphur	13	Lead	90
Magnesia	20	Silver	190
Lime	24	Gold	190
Soda	28	Platina	190
Potash	42	Mercury	167


LEMENTS
W.t
Strontian 46
Barytes 68
Iron 50
Zinc 56
Copper 56
Lead 90
Silver 190
Gold 190
Platina 190
Mercury 167


WATER
AMMONIA
1
2

Five wooden balls, made by Peter Ewart of Manchester c.1810, and used by John Dalton for demonstrating his atomic theory c. 1810–42

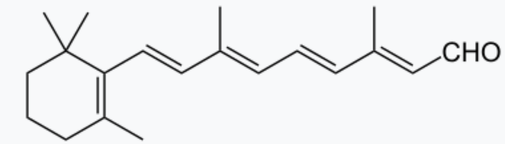
Science Museum Group Collection

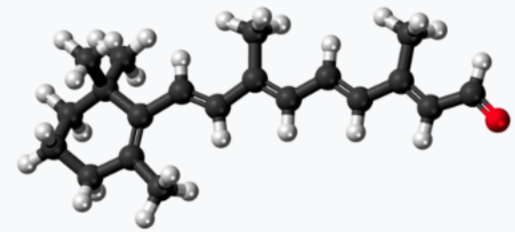


https://en.wikipedia.org/wiki/Vertebrate_visual_opsin

Three-dimensional structure of bovine rhodopsin. The seven transmembrane domains are shown in varying colors. The retinal chromophore is shown in red.
Wikipedia

All-trans-retinal





Names

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Atoms responsible for John Dalton’s colour blindness

https://www.quaker-tapestry.co.uk/2021/03/19/john-dalton-scientist-quaker-and-pioneer/

William Allen

•Wellcome Trust Corporate Archive
•…Direct activities WT/D
•Library and Museum WT/D/1
•Images from the photographic library WT/D/1/20M0007729:

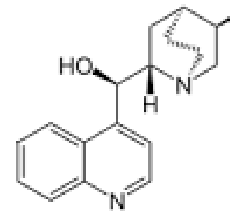
•Interior of the old pharmacy at Plough Court

WT/D/1/20/1/66/62

Plough Court Pharmacy Medicines (1715-1927) [from CursorAI]

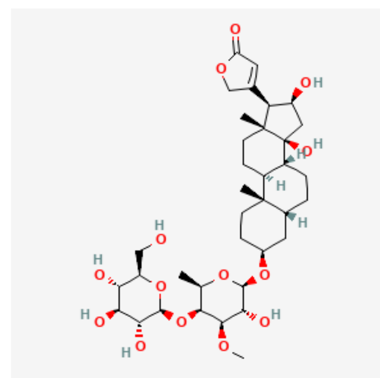
NATURAL:

•**Digitalis (Foxglove)**: For heart conditions
•**Willow Bark**: Pain relief and fever reduction (precursor to aspirin)
•**Quinine powder**: Malaria treatment
•**Chamomile**: Digestive issues and anxiety
•**Peppermint**: **Valerian**: **Rose Water**:
•**Laudanum**: Opium tincture for pain relief and coughs
•**Elixirs, Cordials, bitters**



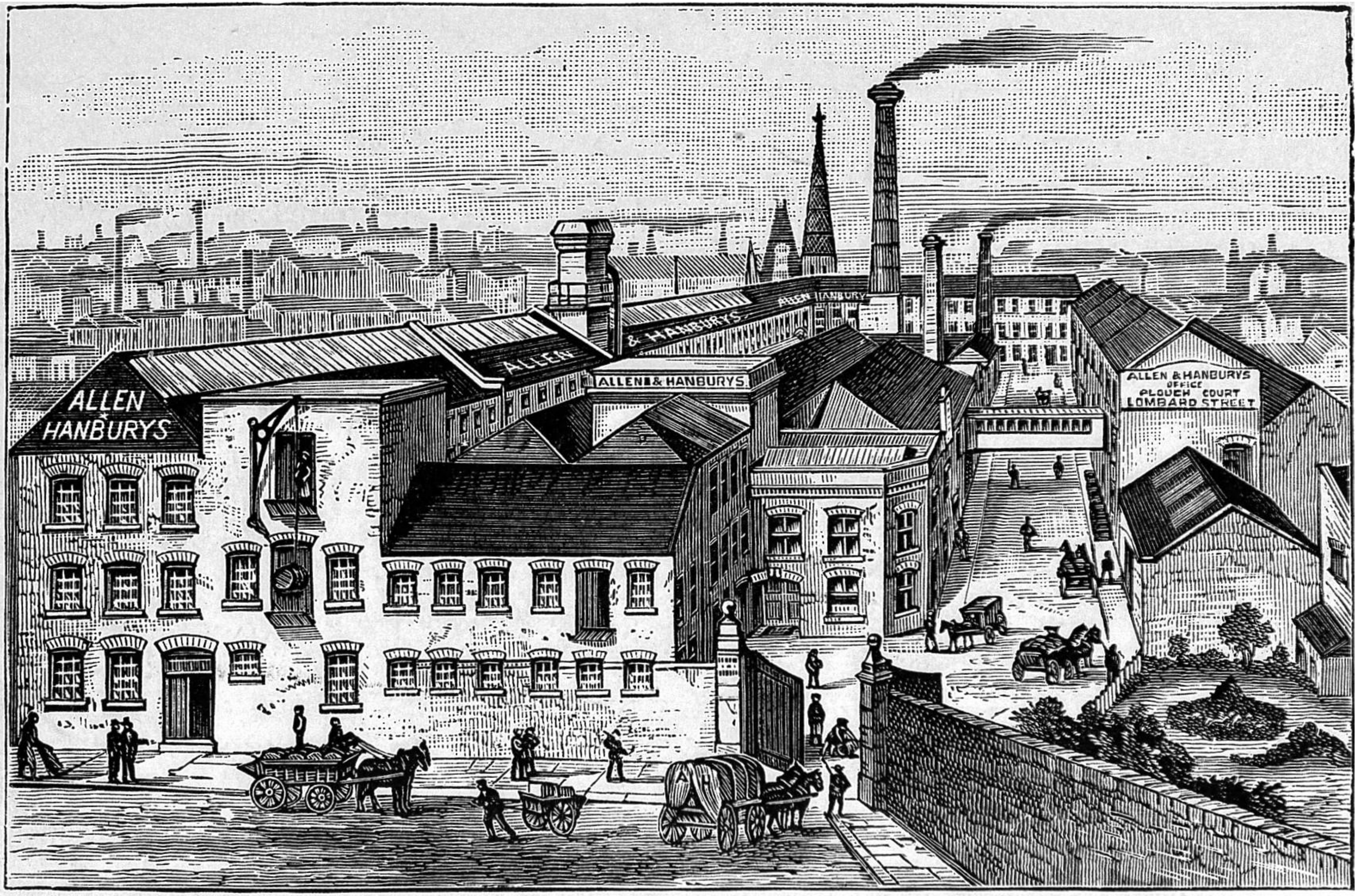
Chemical Compounds

•**Mercury compounds**: For syphilis treatment
•**Calomel (mercurous chloride)**: Purgative
•**Antimony**: Emetic and purgative
•**Arsenic preparations, Lead acetate**
•**Sulfur preparations**: For skin problems
•**Zinc oxide ointments**: For wounds and burns
•**Bottles, Scales, equipment, Mortars and pestles, Distillation equipment**



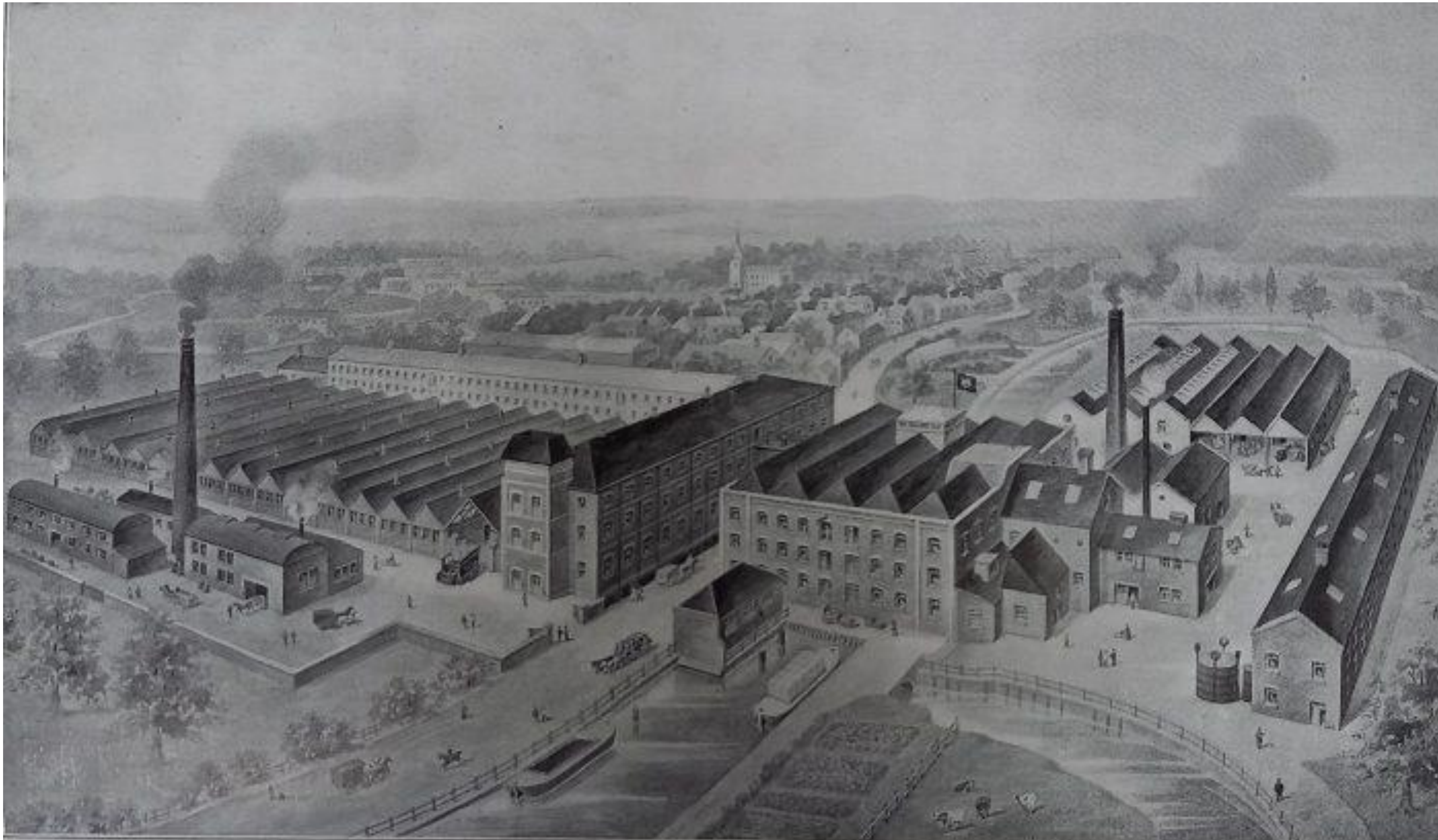
•**Pills and powders**: **Lozenges**:


ALLEN & HANBURYS LTD.



GENERAL VIEW OF THE WORKS AT BETHNAL GREEN.

Creator: Wellcome Library, London
Copyright: Copyrighted work available under Creative Commons Attribution only licence CC BY 4.0 http://creativecommons.org/licenses/by/4.0/



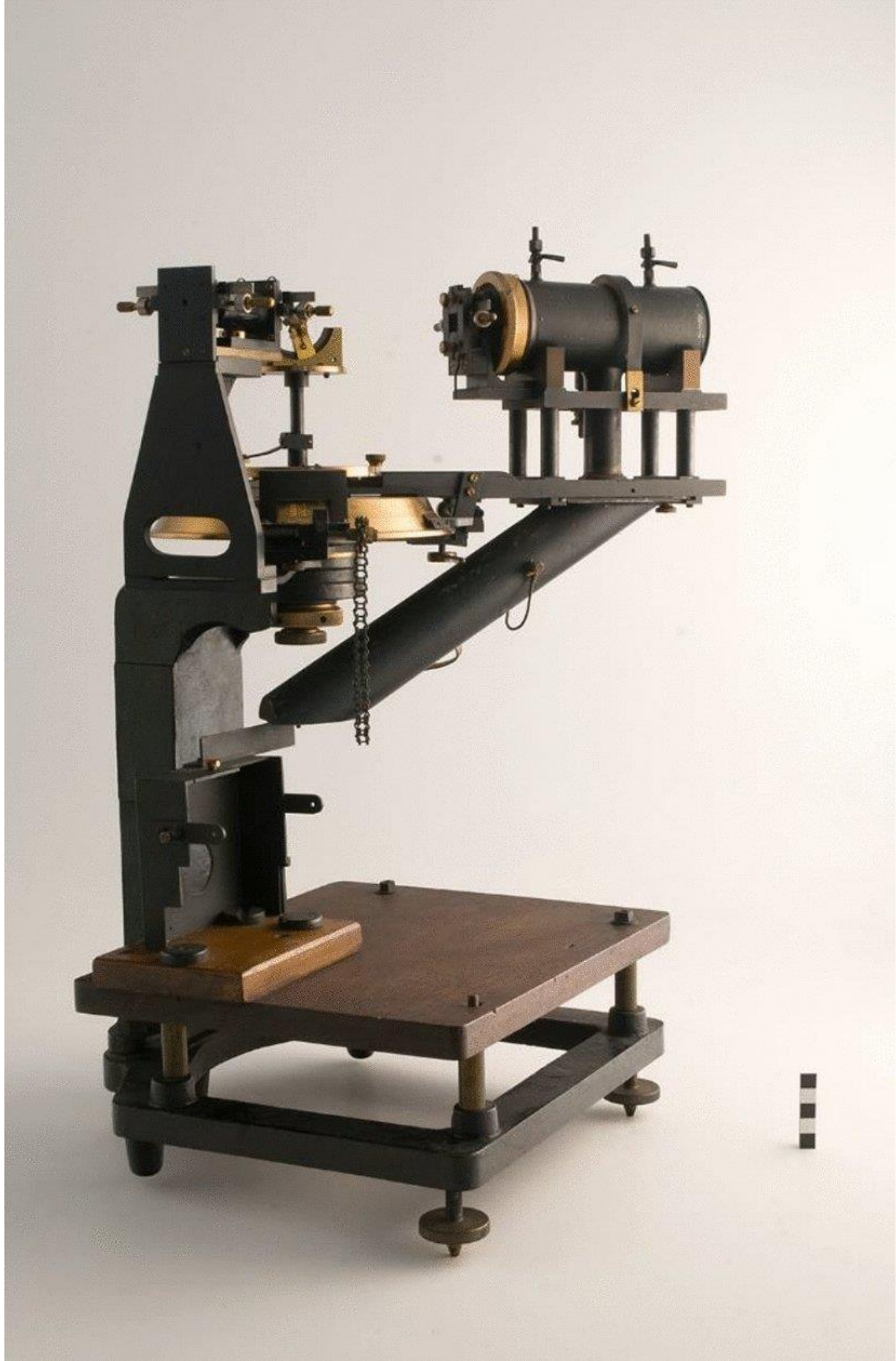
A BIRDSEYE VIEW OF WARE MILLS, HERTFORDSHIRE.

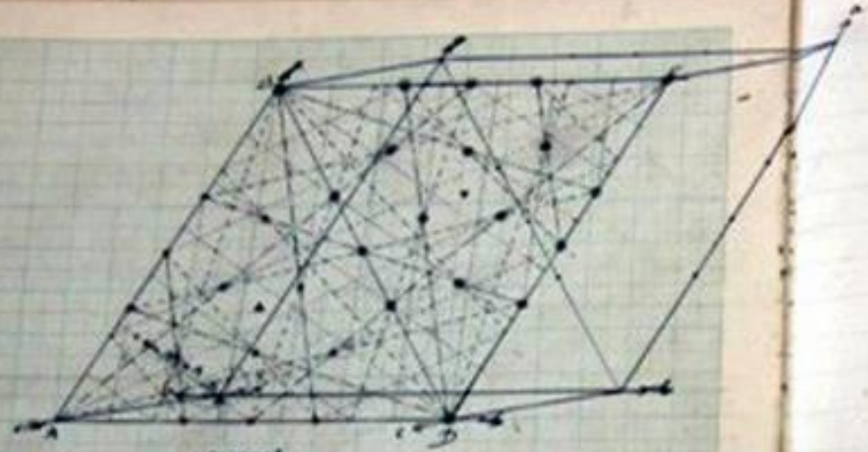
Allen & Hanbury's, Ware showing the factory and mills in 1920

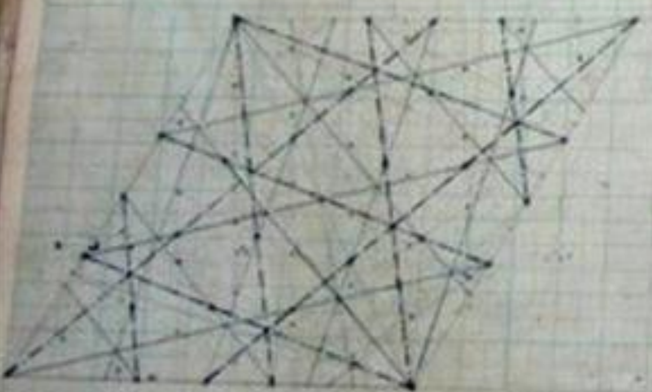
Hertfordshire Archives & Local Studies (Ware.Industries pamphlet file)

Kathleen Lonsdale

Kathleen Lonsdale in her crystallography research laboratory at UCL. Photographer and







$$\sin^2\frac{a}{2} = \frac{\sin(S-B)\sin(S-C)}{\sin B \sin C} \quad \text{or} \quad \cos a = \frac{\cos A + \cos B \cos C}{\sin B \sin C}$$

$A = 100 \wedge 001$ etc. $\quad 2S = A + B + C$

$$\frac{\sin a}{\sin A} = \frac{\sin \beta}{\sin B} = \frac{\sin \gamma}{\sin C} \quad \text{etc.}$$

a <u>[illegible]</u> of A acute etc. (A = ∠ between normals)

In above example $A = 50°30'$ $\quad \alpha = 135°55'$
$B = 80°49'$ $\quad \beta = 47°4'$
$C = 116°19'$ $\quad \gamma = 57°50'$

$$\frac{a}{b} = \frac{\sin 001 \wedge 010 \cdot \sin 010 \wedge 100}{\sin 001 \wedge 100 \cdot \sin 100 \wedge 010} \quad \text{etc.}$$

whence (taking mean of all angles)

$a : b : c = 1.007 : 1 : 0.5964$

Axial ~~spacings~~ lengths calculated from $a = \frac{d_{100} \sin \alpha}{N}$ etc.

where $N^2 = 4 \sin S \sin(S-\alpha)\sin(S-\beta)\sin(S-\gamma)$

$S = \frac{\alpha+\beta+\gamma}{2}$

from above $\log N^2 = \bar{1}.54635$ $\quad \log N = \bar{1}.77317$ $\quad N = .5930$

$d_{100} = 7.838$ $\quad a = 9.179$

$d_{010} = 6.069$ $\quad b = 9.097$

$d_{001} = 3.706$ $\quad c = 5.400$

whence $\quad a : b : c = 1.007 : 1 : 0.5931$

$a . b . c . N . \rho = m . M . m_H$

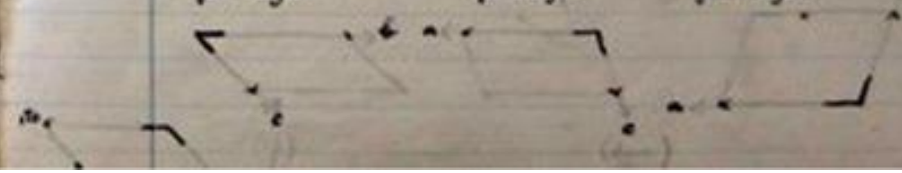
$M = 162.2 \quad m_H = 1.649$

$$\frac{m}{\rho} = 1.00$$

$\rho = 1.060 \quad$ whence $m = 1.060 \quad$ too large

d_{100}, d_{010}, d_{001} <u>repeated</u>, also A, B, C.

looking along a axis from right $\quad$ looking along b from right $\quad$ looking along c from right

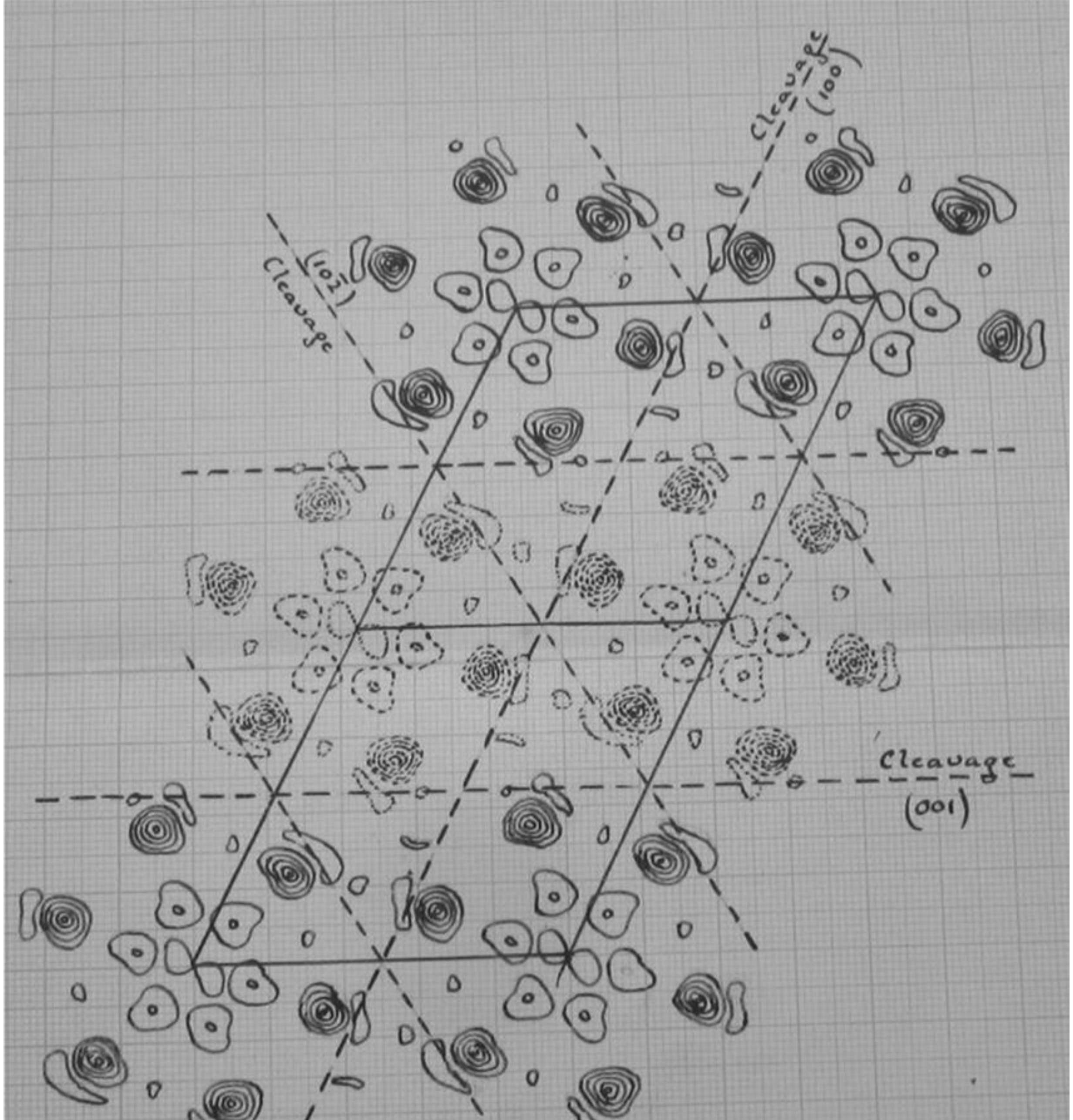


Monoclinic System.

Monoclinic Hemihedry (Domatic).

No.	S.-G.	B.L.	*n*.	Abnormal Spacings.	*p*.	Possible Molecular Symmetry.
3	C_s^1	Γ_m	2	None	2	P.
4	C_s^2	Γ_m	2	(*a.*) $\{hol\}$ halved if h is odd (*b.*) $\{hol\}$ halved if l is odd (*c.*) $\{hol\}$ halved if $(h+l)$ is odd	—	None.
5	C_s^3	Γ_m' (*cf.* Γ_0''')	4	(*a.*) $\{hkl\}$ halved if $(h+k+l)$ is odd	2	P.
		(*cf.* Γ_0')	4	(*b* i.) $\{hkl\}$ halved if $(k+l)$ is odd		
		(*cf.* Γ_0')	4	(*b* ii.) $\{hkl\}$ halved if $(h+k)$ is odd		
		(*cf.* Γ_0'')	8	(*c.*) $\{hkl\}$ halved if $(h+k)$ or $(k+l)$ or $(l+h)$ is odd		
6	C_s^4	Γ_m'	4	(*a.*) (*b* i.) (*b* ii.) Same as C_s^3 (*a.*) (*b* i.) (*b* ii.) respectively; also all $\{hol\}$ halved.	—	None.

International Tables for Crystallography (230 spacegroups), major contributions from Kathleen Lonsdale

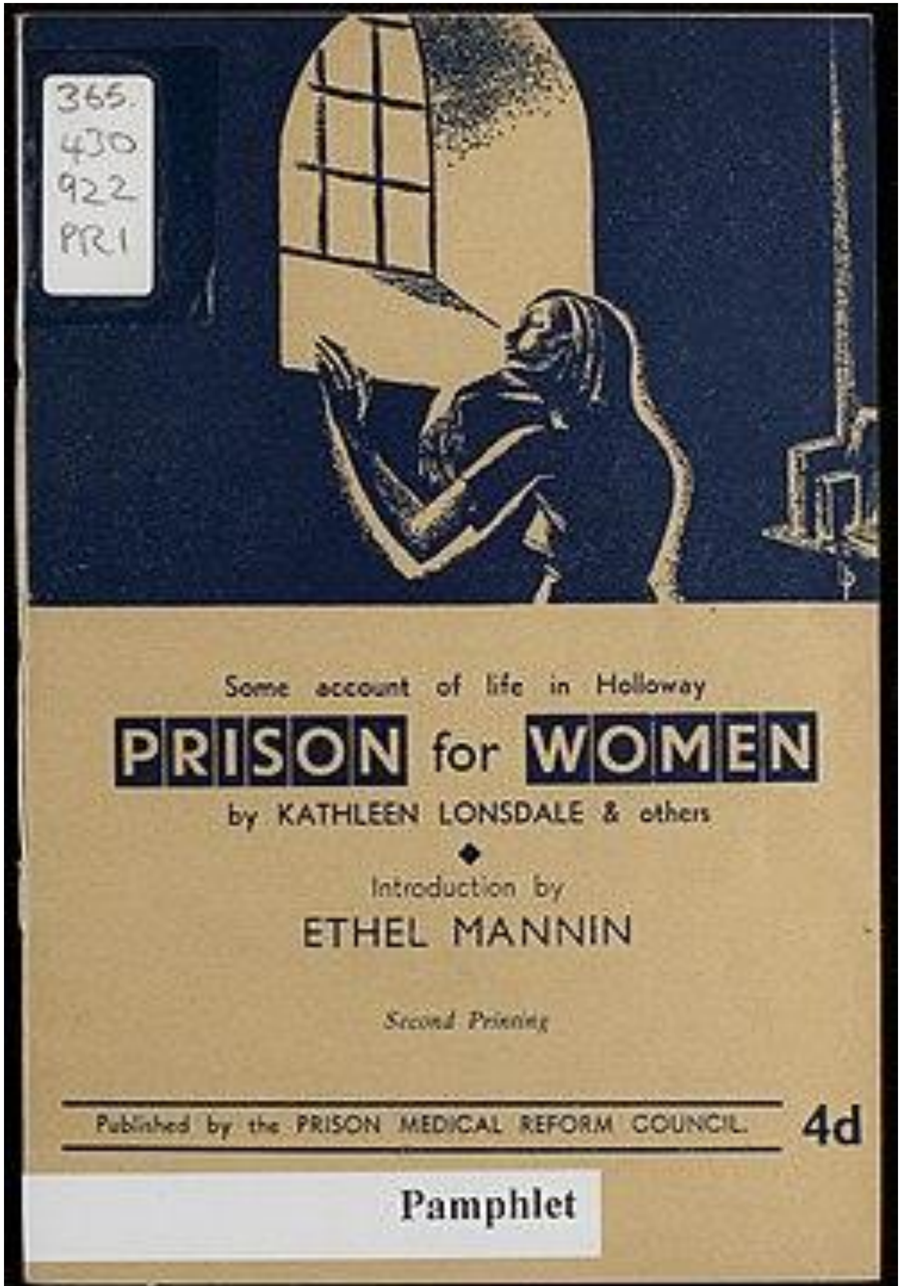


Drawing of the electron density projection on the (010) plane calculated for hexachlorobenzene. Extract from Lonsdale’s notebook labelled ‘1924’ Papers of Dame Kathleen Lonsdale UCL Special Collections. Courtesy: The Lonsdale family

Kathleen Lonsdale

…another baby and I spent most of my time finishing off (by hand, of course, using log tables) the Fourier analysis of hexachlorobenzene. In view of the oft-repeated claims of organic chemists that the crystallographers told them nothing that they did not already known, it may be interesting to reprint a more generous tribute from **C. K. Ingold** (dated 2 November 193 1).

‘Ever so many thanks for the reprint of your wonderful paper on Hexachlorobenzene. I havejust been studying it in the Royal [Society journals] but am very glad to have it in a handy form. **The calculations must have been dreadful but one paper like this brings more certainty into organic chemistry than generations of activity by us professionals**.’



Of her experience in prison, she said,

“What I was not prepared for was the general insanity of an administrative system In which lip service is paid to the idea of segregation and the ideal of reform, when in practice the opportunities for contamination and infection are innumerable and those responsible for re-education practically nil”.

Kathleen joined the Howard League for Penal Reform and frequently visited women in prison as a form of support.

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from Wikimedia. Other quoted material is offered as "personal use or non-commercial".
We suggest you treat this set of slides as CC BY-NC.
(Both Graham and Peter have liberal approaches to copyright (CC BY)

Repository. These slides and associated material are at
https://github.com/petermr/quaker_scientists

Use of AI. CursorAI (2025-09) has been used to gather information, primarily
from Wikipedia. I have checked most of it for accuracy but there may be slight errors

Knowledge Graphs. These have been created by CursorAI and I am continually revising
The protocol. There may be errors